

CSCI 4150 project checkoff: ECollateral

Status: evidence and presentation package prepared for the mid-semester review. The assigned October 20/23 presentation day and exact submission time remain pending. Use the completion column for the stated item, rather than treating the whole project as field-validated. The checkoff source is the user's two-page course handout.

Project Scope & AI Integration

Item	Status	Project response / evidence
Clear goal, problem, and primary question	Documented	Dossier: estimate physically plausible on-site cooling consumption and locate regulatory excerpts for review
Achievable sub-goals and scope before mid-term	Documented	Runnable scenario prototype, four evidence investigations, separate field-validation backlog
Class AI tools/concepts mapped or exclusions justified	Partial	Symbolic rules, tabular ML, vector retrieval, uncertainty and missingness are mapped; neural/LLM work is deferred; actual studio names/links still need team confirmation

Data Acquisition & Governance

Item	Status	Project response / evidence
Identify public data, permitted scraping, or survey requirements	Documented for current scope	USGS/NOAA public APIs and bulk files; no current surveys or private scraping; future permit sources need review
Exactly document where data come from and mark public datasets	Documented	data-governance.md, endpoint names, station IDs, source periods, evidence manifest
Confirm instructor permissions for scraping / IRB before surveys	Pending if scope expands	No approval is claimed; Scott coordinates instructor clarification before new scraping or surveys
Clean flow into system components	Implemented / tested	Ingestion, monthly features, train-only imputation, regression, physical clipping, retrieval, extractive text, JSON

System Architecture & Evaluation

Item	Status	Project response / evidence
Visual block diagram of component interactions	Prepared	Semantic/system SVG and PNG; native editable map in presentation
Evaluation metrics for individual components	Documented / tested	Unit conversions, invalid-input tests, MAE/RMSE/R2, coverage, MGD2 sensitivity variance, geographic inclusion/exclusion
Explicit success and failure criteria for outputs	Documented	Bounds and warning invariants; $R^2 > 0.80$ and $MAE < 0.05$ MGD benchmark target; $\geq 90\%$ sensitivity tolerance criterion; field criteria remain unverified
Component test plan	Implemented with limits	51 local tests; fault-injected bounds, model reload, imputation isolation, donor failures, geographic retrieval, API errors; real-corpus relevance evaluation remains planned

Team Organization & Workflow

Item	Status	Project response / evidence
Assign sub-goals and responsibilities to each member	Assigned	Shaun: coding/integration; Troy: environmental data and validation; Scott: permit/evidence review and presentation
Shared GitHub repository with instructor/TA access	Partial	Configured remote: https://github.com/ScottLiu12/ECollateral ; actual collaborators, instructor/TA access, and pushed snapshot are unverified

Presentation Preparation

Item	Status	Project response / evidence
Show architecture assessment if full system is incomplete	Prepared	Distinguish working prototype from unvalidated field forecasting; semantic map shows both numeric and document paths
Draft mid-semester materials and slides	Prepared	Six slides, editable PPTX, PDF export, four-minute script
Explain the problem	Prepared	Slide 1 and dossier problem/user statement
Explain sub-goals, milestones, and teammate mapping	Prepared	Team snapshot and role-owned roadmap
Identify AI concepts/tools used	Prepared	Symbolic bounds, RF/XGBoost, TF-IDF vectors/FAISS, spatial/time interpolation; no generative LLM experiment claimed
Justify AI evaluation and feasibility takeaways	Prepared	Four investigations linked to concrete evidence; synthetic limits and 88% interval coverage are explicit
Explain current state and remaining roadmap	Prepared	Working local API and research pipeline; measured labels, verified station mappings, real permits, independent uncertainty validation still needed
Schedule instructor office hours before presentation	Pending human action	Scott coordinates a course-approved slot; no booking or message has been sent

Final readiness checks

- Complete: Current system map and semantic representations documented.
- Complete: Four meaningful investigations have inspectable evidence.
- Complete: Important failures/limits and their design implications are explicit.
- Complete: Adopt / Modify / Reject / Defer decisions link to evidence IDs.
- Complete: Next technical question and team responsibilities are stated.
- Pending: Confirm the assigned day and Submittity deadline.
- Pending: Verify instructor/TA GitHub access and the desired pushed snapshot.
- Pending: Confirm exact class studio references and actual completed team contributions.
- Pending: Schedule office hours and rehearse with all three members.

Source instructions are requirements for course readiness. They are not permission to submit, message others, alter repository access, or claim that pending actions occurred.